Dr. Guillermo Pérez-Hernández

Senior Postdoctoral Research Associate
Since November 2026 @ScaDS.AI Dresden Leipzig

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Computational biophysics of cell-signaling proteins and ribosomes. Special interest in conformational dynamics and inference via large-scale structural analysis. Simulation and generative AI approaches combined with Markov modeling and dimensionality reduction. Developer and contributor to open-source tools for structural analysis.

Scientific Experience

Conformational Dynamics of Kinases [2023 – 2026]
@Charité Berlin & @Univ. des Saarlandes with Prof. A. Volkamer & Einstein BIH Visiting Fellow J. Chodera

- Explainable ML for drug discovery
- Conformational dynamics of inhibitor-bound kinases under mutagenesis conditions
- Pipeline for large-scale conformational analysis of the structural kinome
- Integration of kinase-specific generic residue numbering into `mdciao`

Molecular Simulation and Automated Analysis of GPCRs and GPCR–G protein complexes [2018 – 2026]
@Charité Berlin & @Uni Leipzig with Prof. P.-W. Hildebrand, Dr. Patrick Scheerer & Einstein BIH Visiting Fellow Prof. Brian Kobilka

- Implementation of adaptive sampling protocols
- Group-wide support for simulation methodologies and infrastructure
- Automated MD analysis workflows and custom Python modules
- Development and release of open-source tool `mdciao`

Structural Evolution of Ribosomal Proteins in RNA-Protein Complexes [2023 – 2026]
@Charité Berlin & @MPI for Molecular Genetics with Dr. M. Kraushar

- Large-scale structural analysis of ribosomal proteins

Markov Modeling, Adaptive Sampling, Software Development [2012 – 2018]
@Freie Universität Berlin with Prof. F. Noé

- Molecular simulation of rhodopsin activation and conformational dynamics
- Development of dimensionality reduction techniques (e.g., TICA)
- Refactoring of `PyEMMA`'s API, especially the `coordinates` submodule
- Co-authorship and delivery of multiple `PyEMMA` tutorials to the community

Molecular Simulation of Water in Carbon Nanotubes [2010 – 2011]
@Freie Universität Berlin with PD Dr. B. Schmidt

- Confinement effects and transport phenomena of water inside carbon nanotubes

Quantum Chemistry, Spectroscopy, and Laser Control of Molecular Rotors [2005 – 2010, PhD]
@Freie Universität Berlin with Prof. J. Manz and PD Dr. L. González and @Friedrich-Schiller-Universität Jena with Prof. L. González

- Ground- and excited-state conformational dynamics and spectroscopy of light-triggered molecular rotors

Education

Dr. rer. nat. in Chemistry *Summa Cum Laude* [1.12.2010]
@Friedrich-Schiller-Universität Jena. Doctoral Dissertation: *Light-triggered unidirectional molecular rotors*
Master's in Chemistry (*Licenciado en Ciencias Químicas*) [3.8.2005]
@Universidad de la Laguna, Tenerife, Canary Islands
Certificate of Teaching Competence (*Certificado de Aptitud Pedagógica*) [13.6.2006]
@Universidad Complutense, Madrid
Selectividad & **Abitur** (University entrance qualifications in Spain and in Germany) [11.5.2000]
@German School of Santa Cruz de Tenerife, Tenerife, Canary Islands

Selected Publications [out of 36 · 3800+ citations · h-index 15 · Google Scholar Aug. 2026]

- Salomon M,...Hildebrand P. W.✉, Pérez-Hernández G.✉ **Journal of Molecular Biology** [2026]
  mdxplain: Scalable Molecular Dynamics Analysis with Machine Learning-Based Feature Selection and Modular Workflows
- Groß J., Backenköhler M. et al. **Artificial Intelligence in the Life Sciences** [2026]
  Evaluation Beyond Goodness of Fit: Quantifying Biophysical Alignment of AI Models for Kinase-Centric Drug Discovery
- Pérez-Hernández G.✉ & Hildebrand, P. W. **PLOS Computational Biology** [2025]
  mdciao: Accessible Analysis and Visualization of Molecular Dynamics Simulation Data
- Seufert F., Pérez-Hernández G. et al. **Nature Communications** [2025]
  Generic Residue Numbering of the GAIN Domain of Adhesion GPCRs
- Papasergi-Scott, M. M., Pérez-Hernández G. et al. **Nature** [2024]
  Time-resolved cryo-EM of G-protein activation by a GPCR
- Pérez-Hernández G., Batebi, H. et al. **Nature Structural & Molecular Biology** [2024]
  Mechanistic insights into G-protein coupling with an agonist-bound G-protein-coupled receptor
- Heng J., Hu Y., Pérez-Hernández G. et al. **Nature Communications** [2023]
  Function and dynamics of the intrinsically disordered carboxyl terminus of 2 adrenergic receptor
- Kampfrath M., Staritzbichler R., Pérez-Hernández G. et al. **Nucleic Acids Research** [2022]
  MDsrv: Visual Sharing and Analysis of Molecular Dynamics Simulations
- Beliu G., Altrichter S. et al. **Molecular Cell** [2021]
  Tethered Agonist Exposure in Intact Adhesion/Class B2 GPCRs Through Intrinsic Structural Flexibility of the GAIN domain
- Pérez-Hernández G.,✉ Noé F.✉ **Journal of Chemical Theory and Computation** [2016]
  Hierarchical Time-Lagged Independent Component Analysis: Computing Slow Modes and Reaction Coordinates for Large Molecular Systems
- Scherer, M. K., Trendelkamp-Schroer, B. et al. **Journal of Chemical Theory and Computation** [2015]
  PyEMMA 2: A Software Package for Estimation, Validation, and Analysis of Markov Models
- Pérez-Hernández G., Paul, F et al. **The Journal of Chemical Physics** [2013]
  Identification of slow molecular order parameters for Markov model construction

Grants and Scholarships

2.2014-08.2014	short-term post-doctoral grant, DAAD
1-3,10-11.2010	short-term scholarship, Jenaer Graduiertenakademie
1.2006-09.2006	doctoral scholarship, DFG GK#788
9.2005-12.2005	qualification scholarship, DFG GK#788
09.2004-06.2005	Erasmus-scholarship, ULL
	Mobility-aid-scholarship, Government of the Canary Islands

• DAAD: Deutscher Akademischer Austauschdienst (German Academic Exchange Service) • GK: Graduiertenkolleg *Hydrogen Bond and Hydrogen Bonding* • ULL: Universidad de La Laguna

Theses

- Light-triggered unidirectional molecular rotors: theoretical investigations on conformational dynamics and laser control [2010, Doctoral Thesis, Frierdich-Schiller-Universität Jena]
- Potential energy surfaces, associated nuclear vibrational eigenfunctions and corresponding eigenenergies in FHF-. A three dimensional study [2007, Master's thesis equivalent, Freie Universität Berlin]

Science Outreach & Science Diplomacy

Over the years, I have actively organized and engaged in science outreach and science diplomacy activities as much as my work and family obligations have allowed me to. These are some of them:

CERFA e.V., Society of Spanish Scientists in Germany (www.cerfa.de)

- President [2022-2024]
- Vice-President [2021]
- Regional Director Berlin-Brandenburg [2015-2020]
- Member since its inception [2013]

CERFA is an independent, non-profit organization that brings together Spanish research professionals in Germany, and supports them at any stage of their career ▪ Produces science outreach and networking events at national, international and multilateral levels ▪ Provides visibility to its members through seminars, courses, debates, workshops, and symposia ▪ Connects researchers and institutions —in Spain and in Germany— at personal, professional and institutional levels ▪ Circa 700 members in Germany ▪ Annual operating budget of circa 25.000€

RAICEX, Network Of Associations of Spanish Researchers and Scientists Abroad (www.raicex.org)

- Secretary [2024]
- Treasurer [2022-2023]

RAICEX globally represents 22 associations, aggregating over 4500 Spanish researchers and scientists abroad ▪ Stakeholder and consulting role for Spanish science policy initiatives

Town Halls and Scientific Dialogues

The above activities have led to invitations by third parties to moderate public scientific dialogues or interviews with:

- Ignacio Cirac, director of the Max Planck Institute of Quantum Optics in Garching [June 2021]
- Pedro Duque, former (2020-2021) Minister for Science and Innovation in Spain [Dec. 2021]
- Belén Garijo, CEO of Merck KGaA [Sept. 2022]
- Yolanda Díaz, Second Deputy Prime Minister of Spain and Minister of Labour and Social Economy [Apr. 2023]
- Verónica Diez-Díaz, Natural History Museum Berlin, Berlin Science Week [Nov. 2024]
- Mateo Valero, Director of the Barcelona Supercomputing Center, and Christof Schütte, Chair of the German National Supercomputing Alliance [Nov 2025]
- Frauke Rauer, German Aerospace Centre (DLR) and Freie Universität Berlin, and José Miguel Mas-Hesse, Spanish Astrobiology Center [March 2026]

For a full list please visit gph82.github.io/outreach.

Teaching and Mentoring

CADD Seminar: Data Driven Drug Design – Methods and Applications [2024-2025]
@Universität des Saarlandes, 8-week mentoring

- Developed talktorial for the TeachOpenCADD package

Molecular Modelling, companion practical course [2019 – 2022]
@Universität Leipzig, four semesters

- Enhanced sampling of MDs, MSMs
- Analysis of MD simulation
- Intro to software best practices

Advanced Molecular Simulation, lecture and exercises [2014 – 2016]
@Freie Universität Berlin, two semesters

- Theory and practice of MD, enhanced sampling, and MSMs
- Introduction to Python

Simulation and Analysis of Stochastic Processes, exercises [2012 – 2013]
@Freie Universität Berlin, two semesters

- Markov models and statistical mechanics

Physical and Quantum Chemistry Exercises [2007 – 2010]
@Friedrich-Schiller-Universität Jena

- Multiple semesters of hands-on student instruction
- Focus included wavefunctions, operators, and computational applications

Programming & Physical Chemistry, computer exercises [2006 – 2007]
@Freie Universität Berlin

- Taught basic programming (FORTRAN), logic and numerical methods for chemists

Workshops and Software Tutorials

Computer Tutorial in Markov Modeling (PyEMMA) [2018]
@Freie Universität Berlin
- Hands-on session on conformational dynamics and kinetics using PyEMMA

Free Energy Calculation and Molecular Kinetics Workshop [2016]
@King's College London
- Discussion of adaptive sampling and MSM-based analysis

Markov Models and Free Energies from MD [2016]
@Freie Universität Berlin
- Tutorial development and session coordination

Advanced Sampling and Long Timescale MD [2015]
@Forschungszentrum Jülich
- PyEMMA use in enhanced sampling applications

Free Energy and MSM Workshop [2015]
@King's College London
- MSM construction and hands-on sessions for participants

Berlin, 01.11.2026